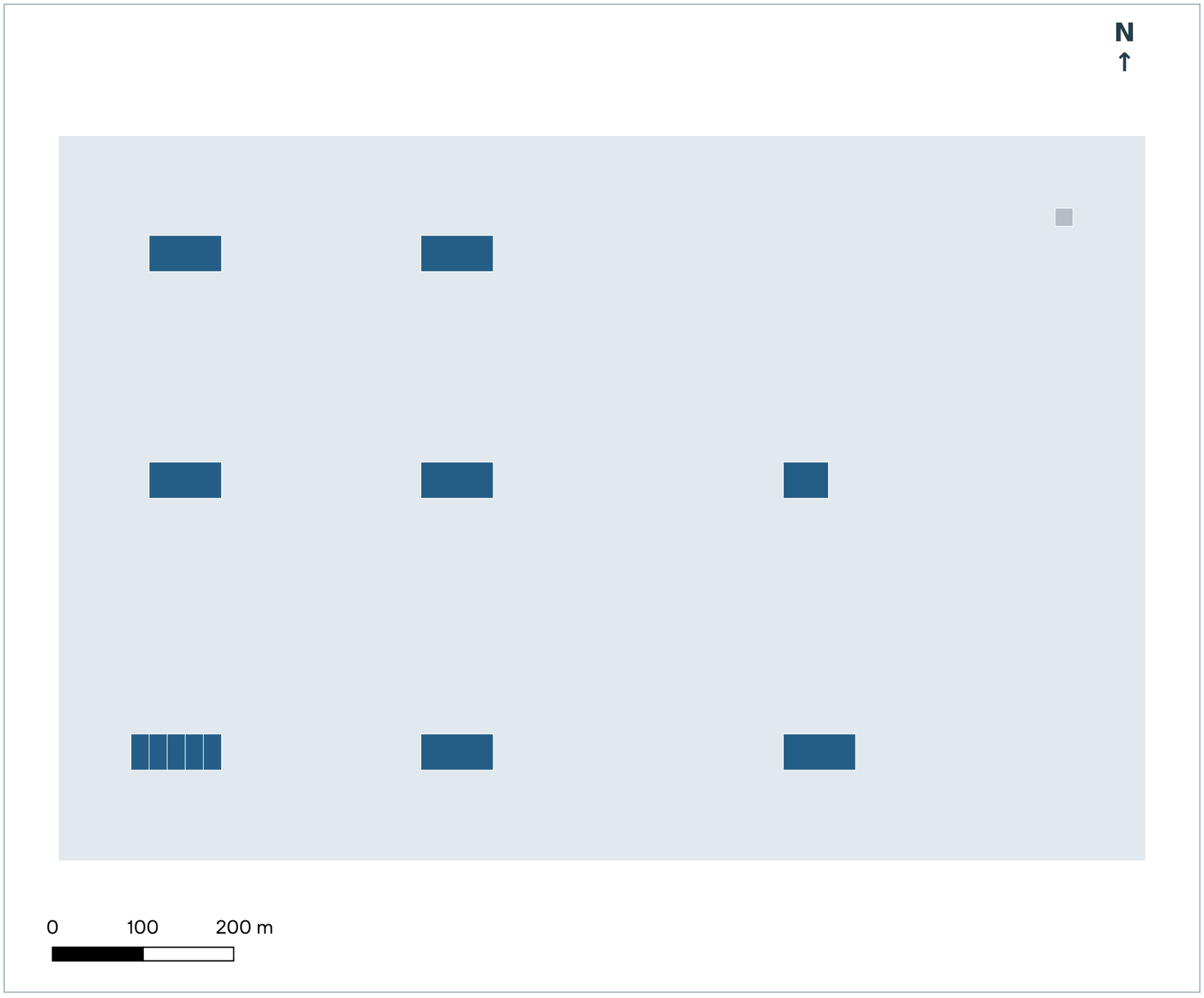
SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

01 · Flood intervention need

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Intervention screen · depth ≥ threshold
- Depth below threshold
- Unknown / NoData

READ THIS MAP

Depth ≥10 cm is the initial user-defined intervention screen. The threshold is configurable.

NoData is unknown, never dry. Upstream source-control options are not removed by this map.

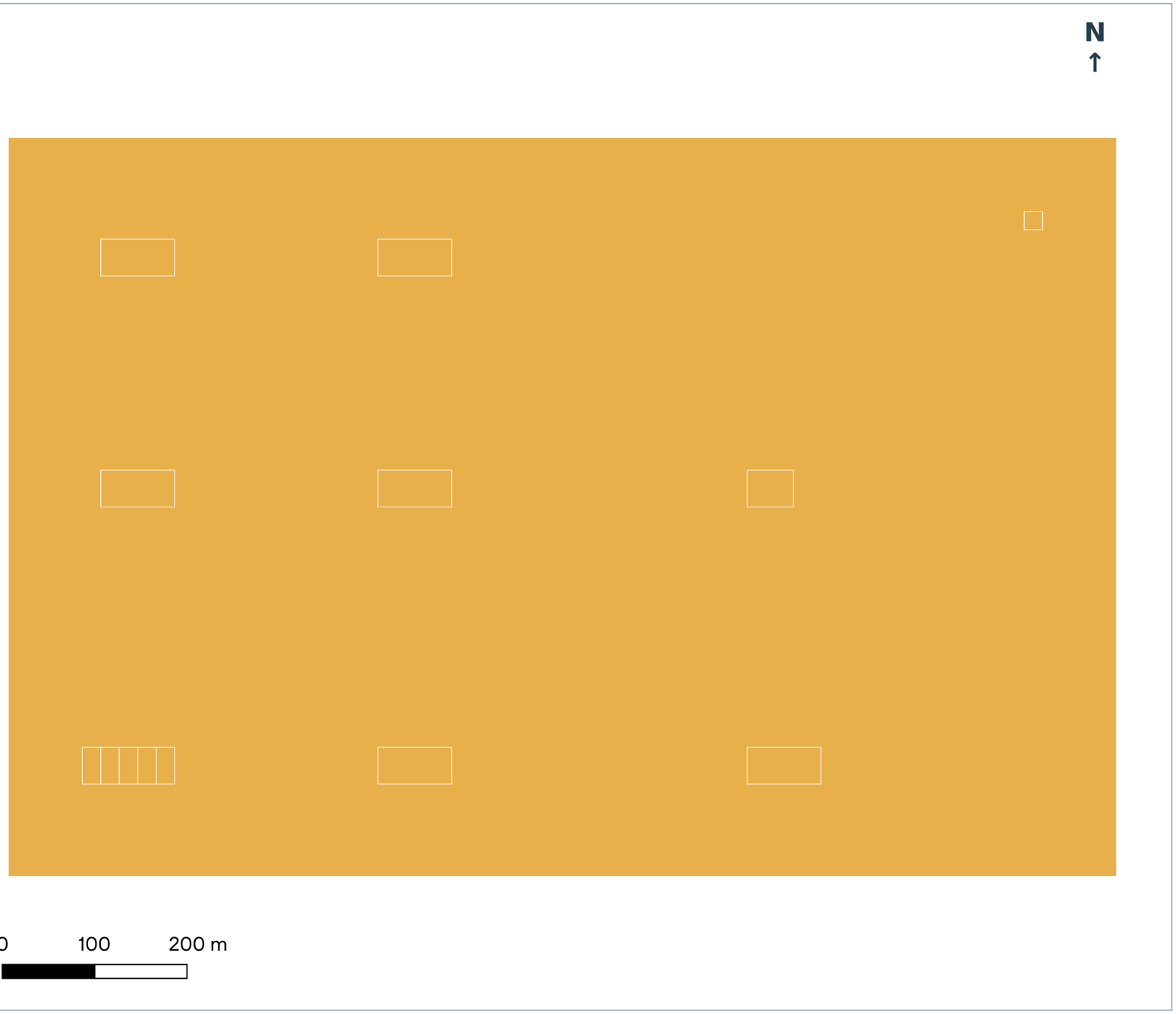
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

02 · Preliminary infiltration pathway

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

This is a pathway-risk screen.
Known prohibitions remain FAIL.
Missing evidence is CONDITIONAL.

Final separation needs a specific module base and a design groundwater state on one datum.

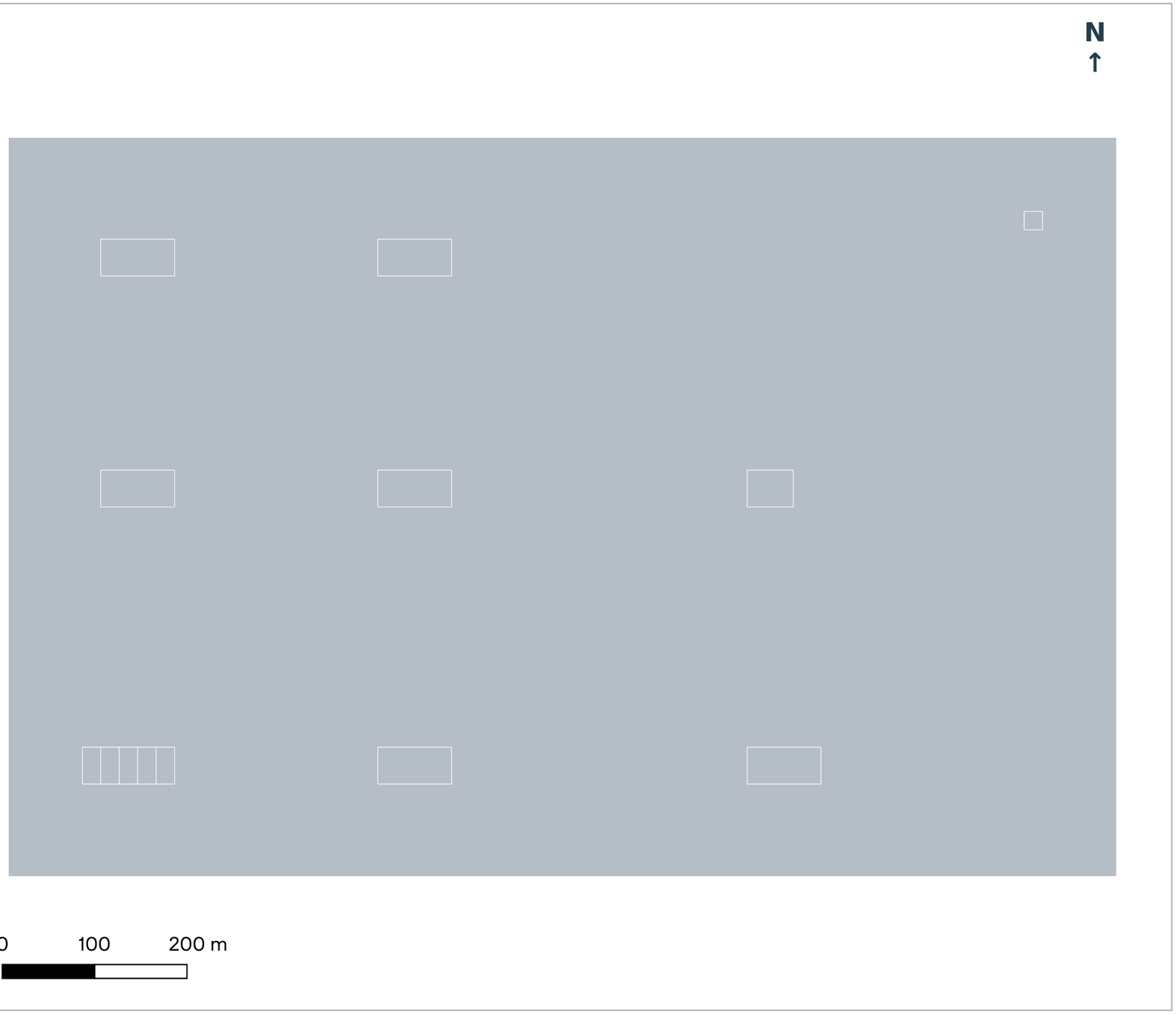
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

03 · Land ownership and delivery

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Unknown ownership

READ THIS MAP

Ownership is intersected by area.
Mixed units retain area shares.
Missing coverage stays unknown.
Overlapping parcels are flagged.

Private land may be feasible
with the necessary agreement.

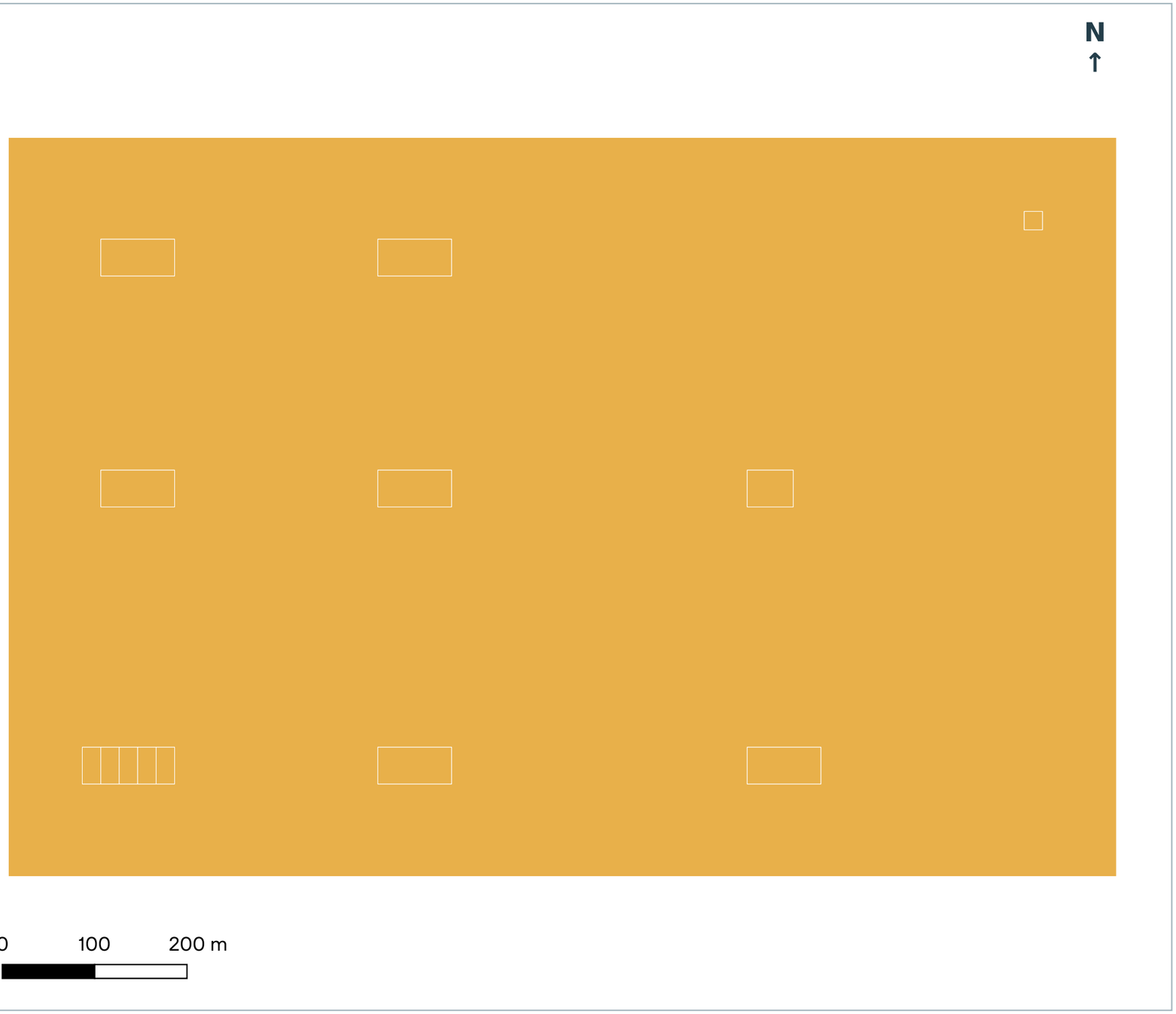
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Green roof

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

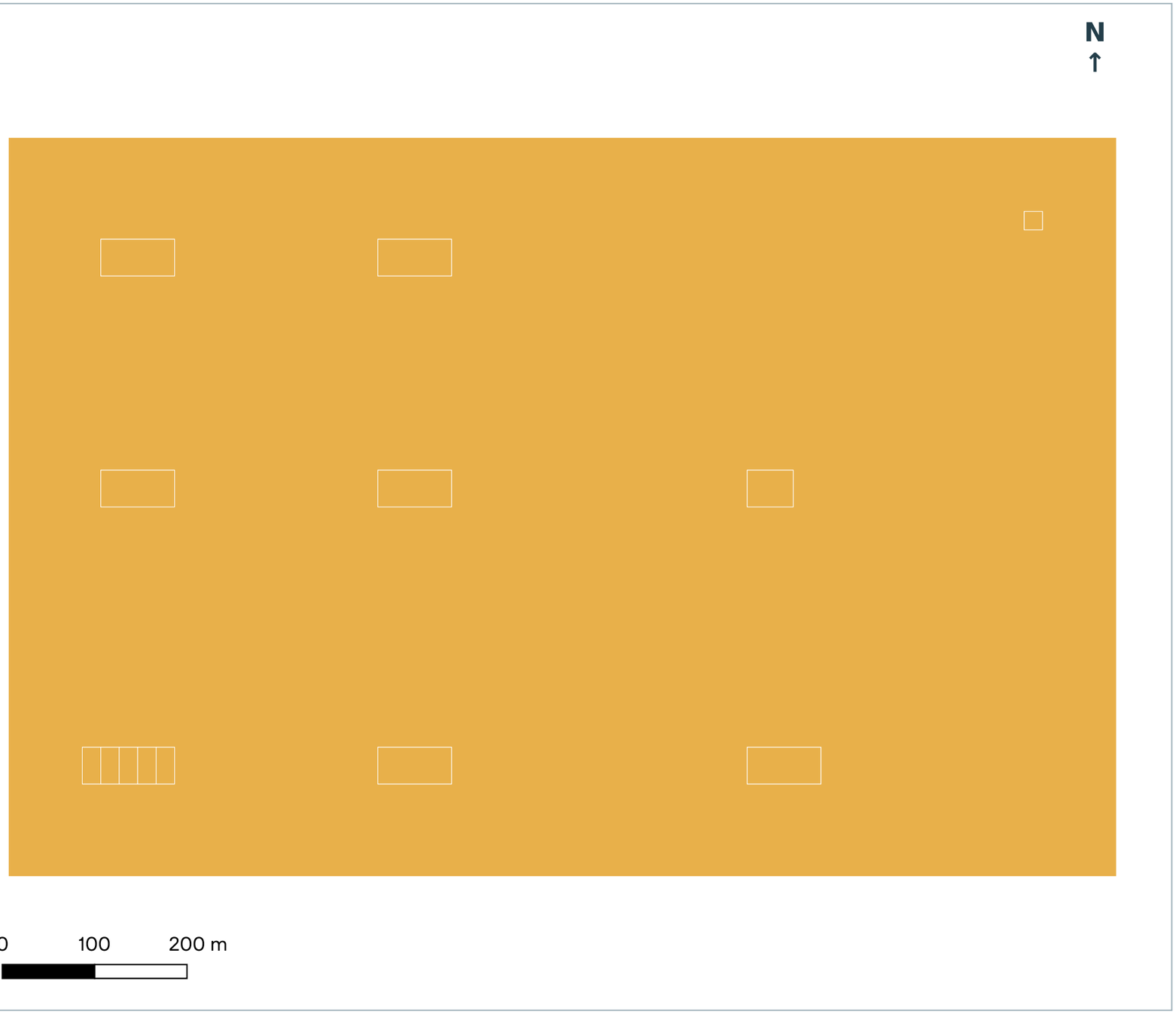
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Blue-green / retention roof

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

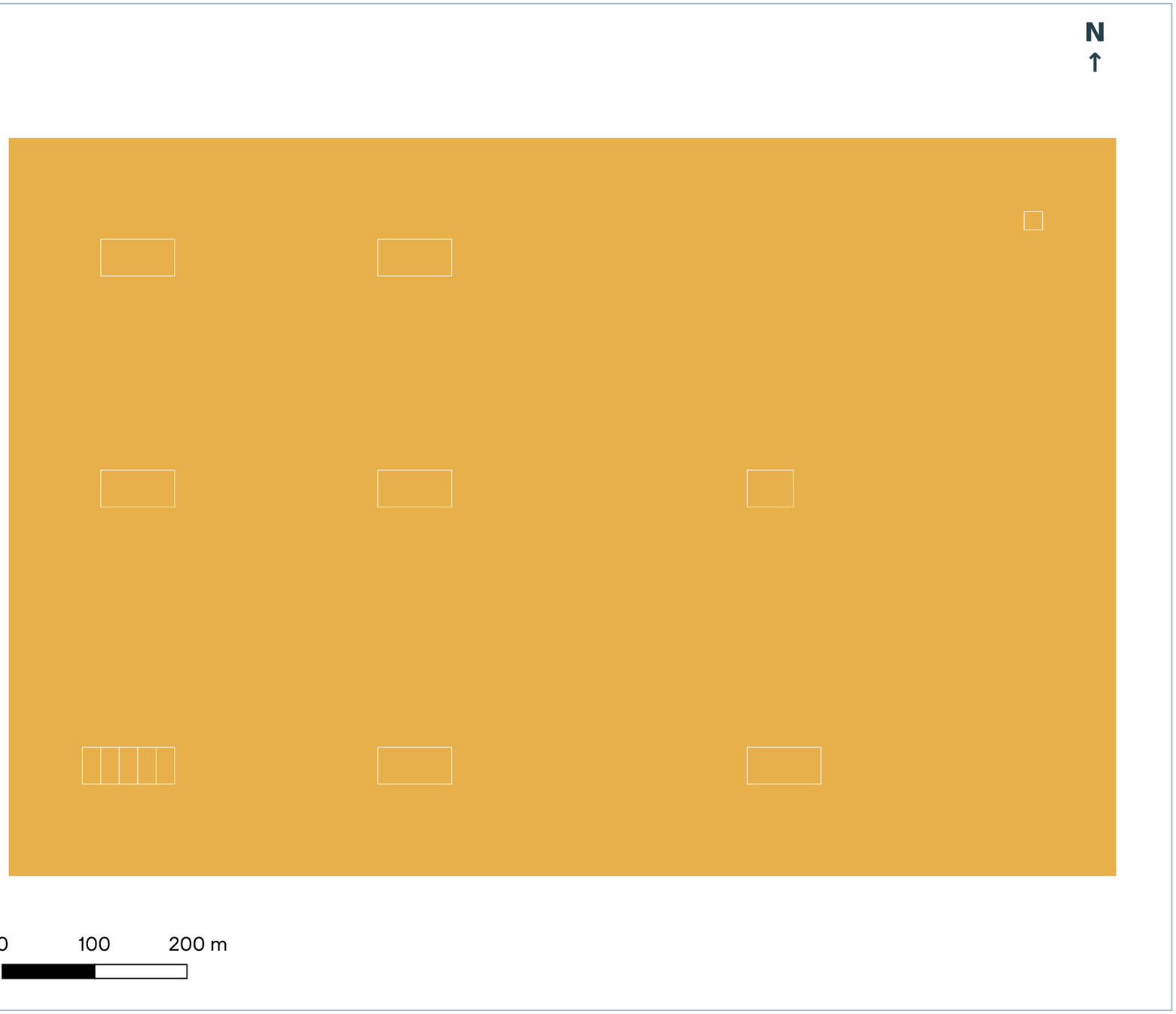
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Living-soil surface infiltration

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

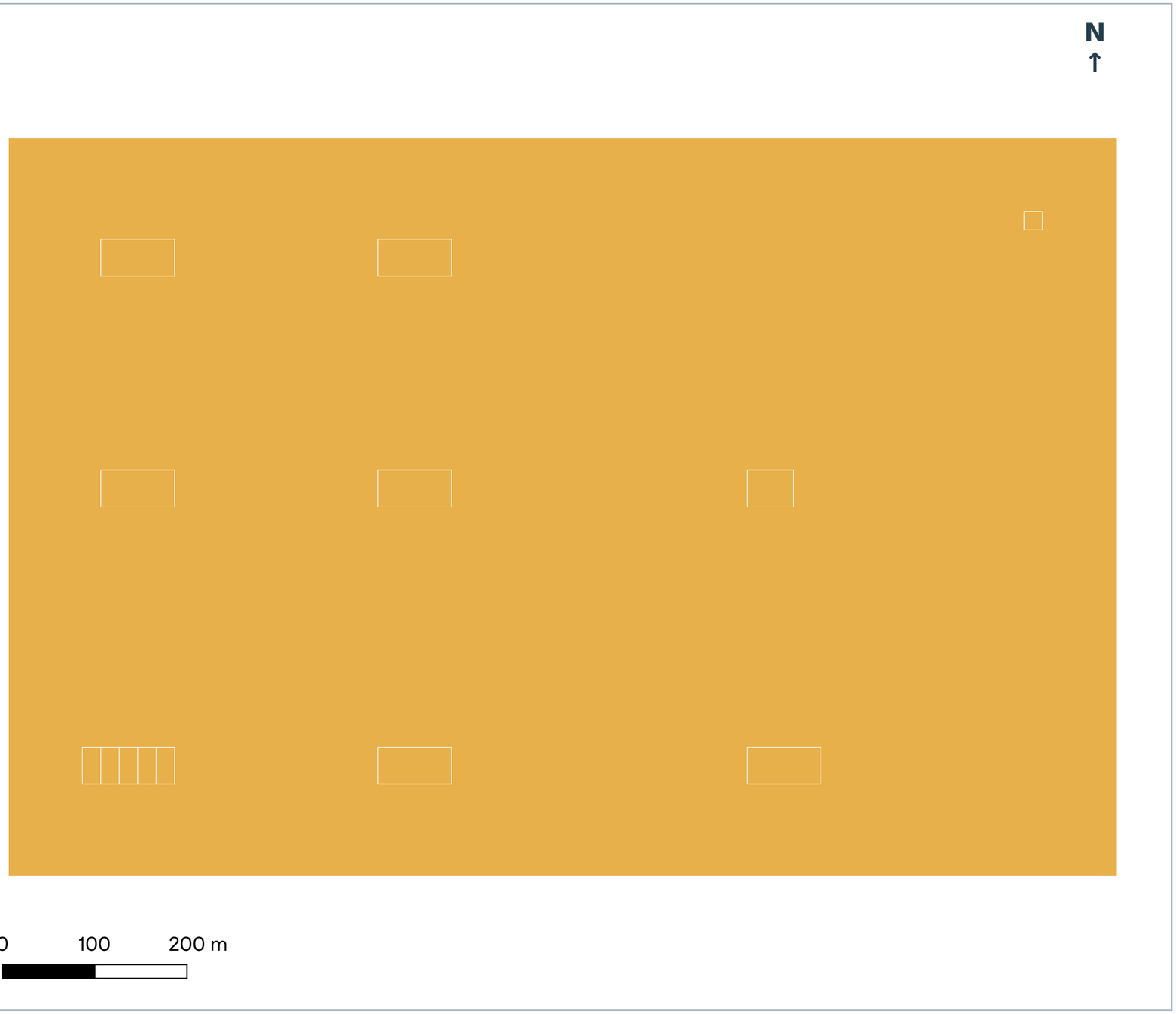
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Vegetated swale / infiltration basin

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated.
FAIL takes precedence over
conditional and admissible.

Each feature stores the reasons,
rule IDs, references, evidence
and required closure actions.

Admissible means screening only.

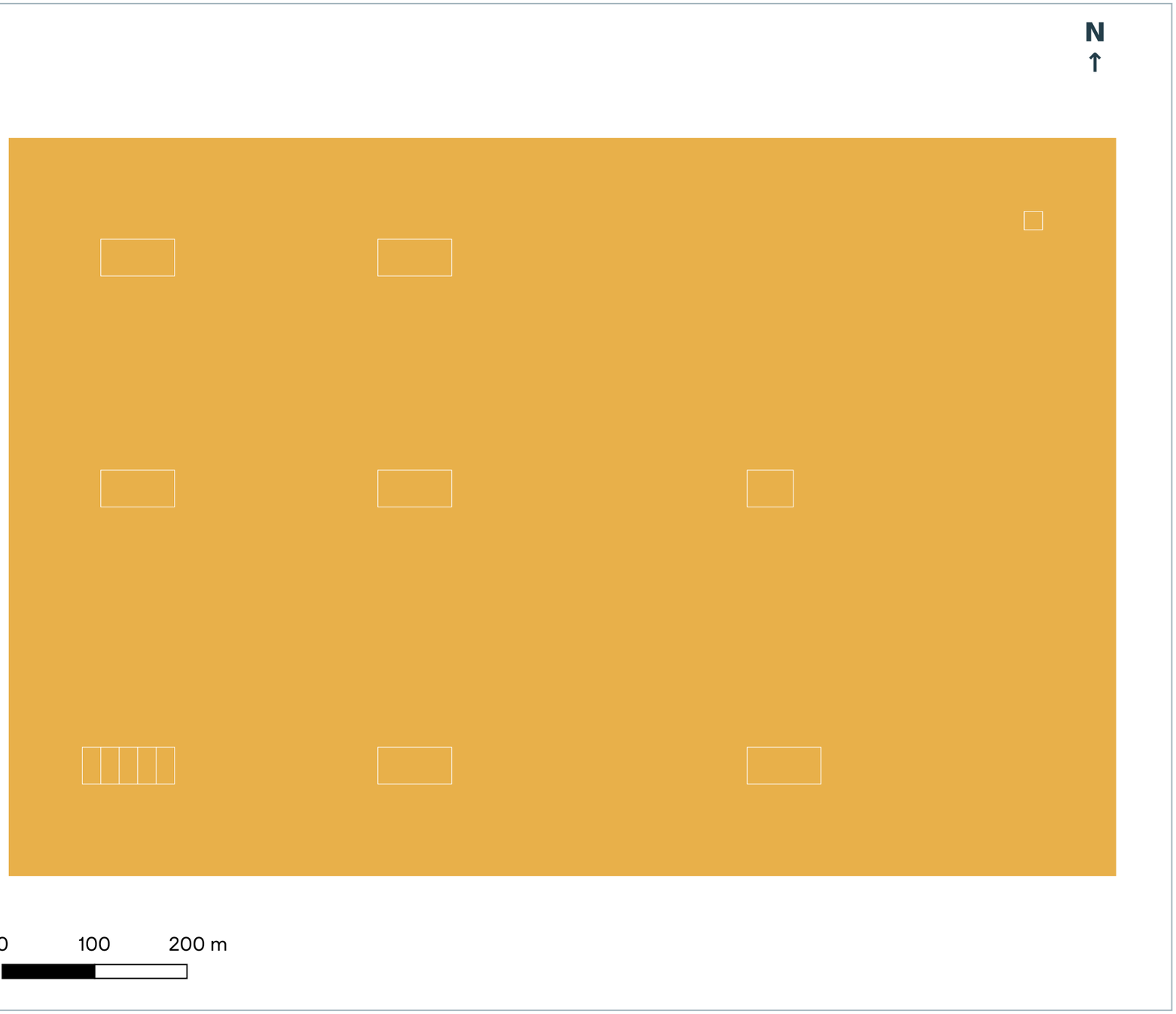
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Rain garden · infiltration

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated.
FAIL takes precedence over
conditional and admissible.

Each feature stores the reasons,
rule IDs, references, evidence
and required closure actions.

Admissible means screening only.

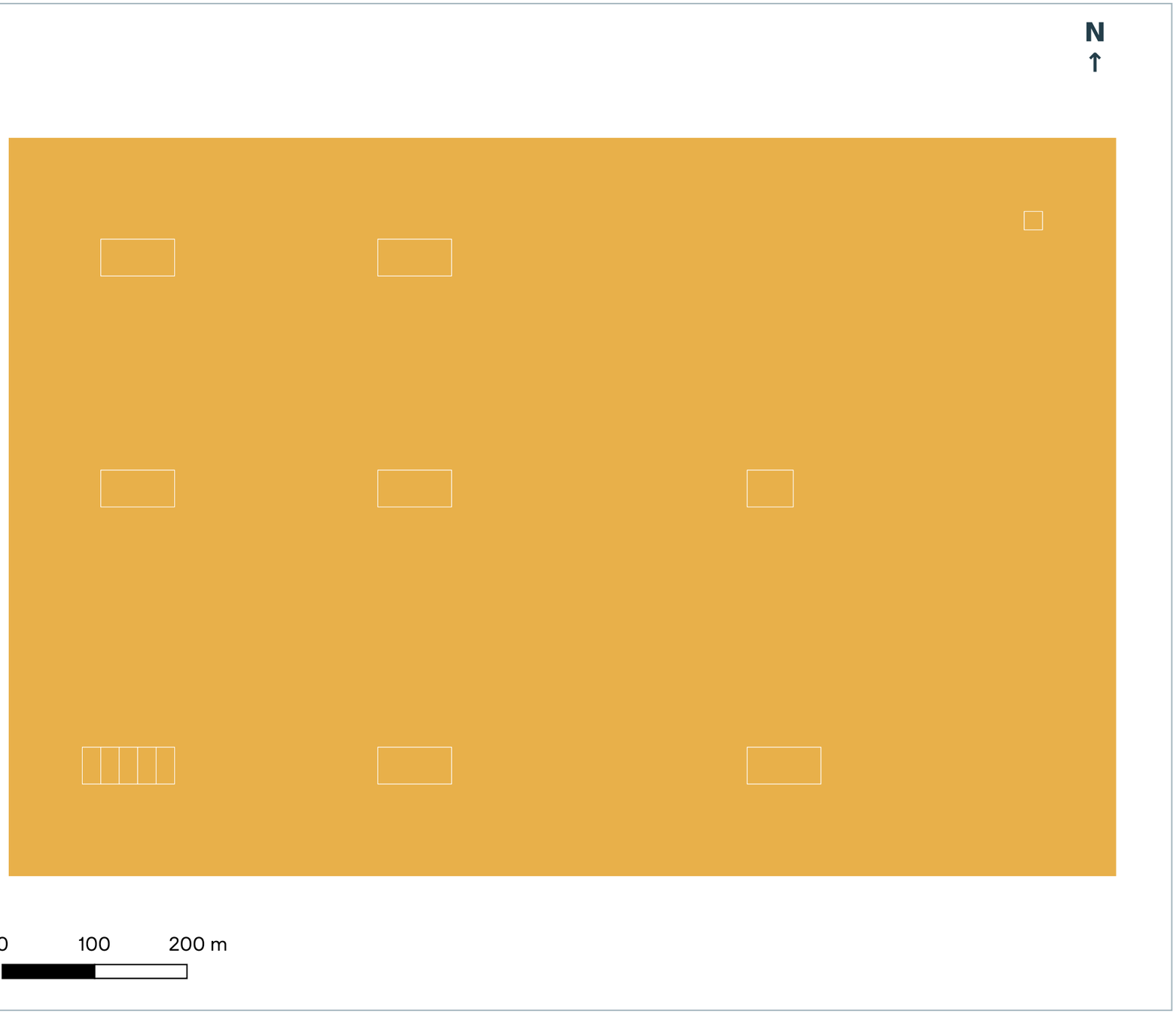
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Rain garden · lined / controlled outlet

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

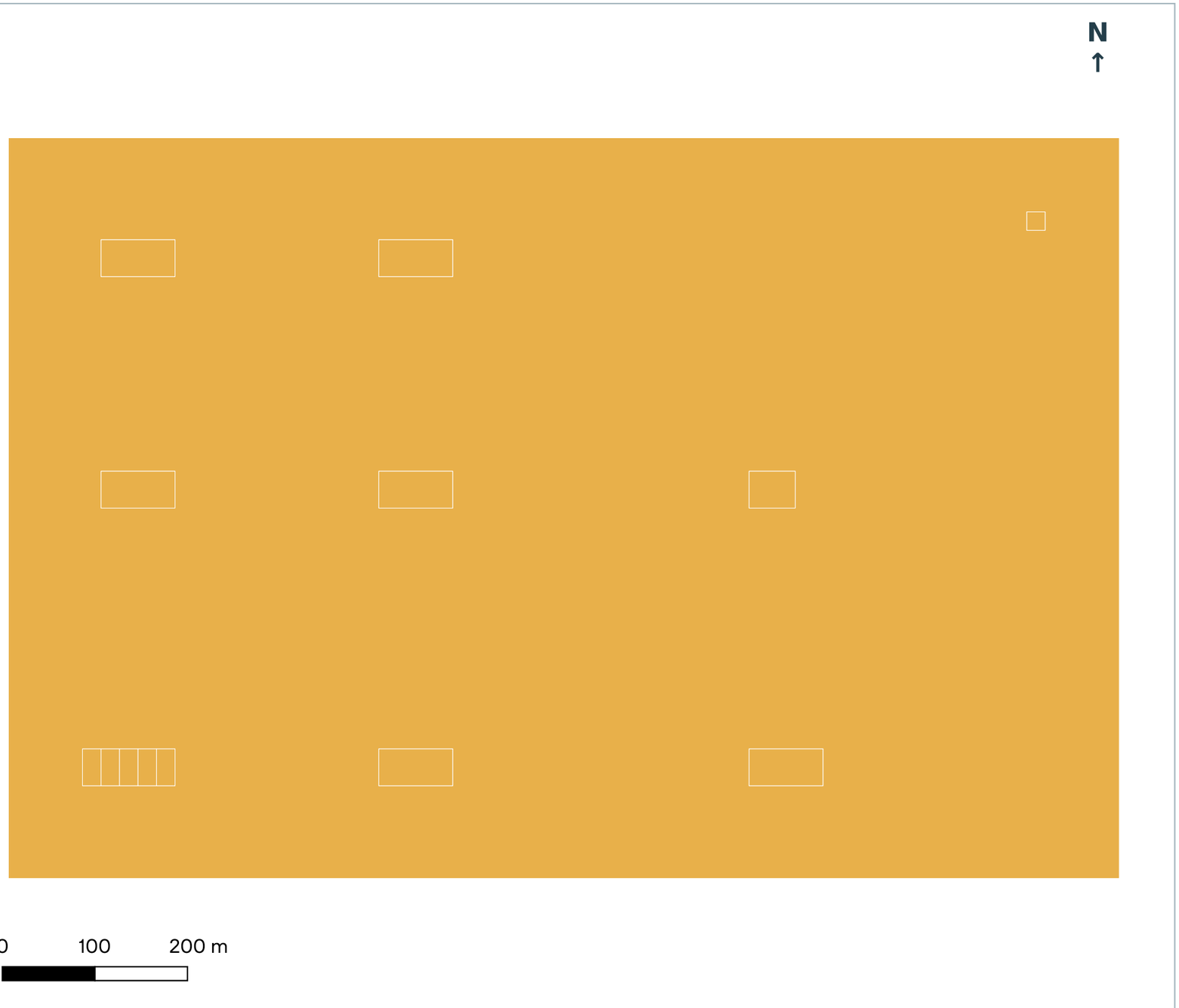
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Tree pit / trench · infiltration

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

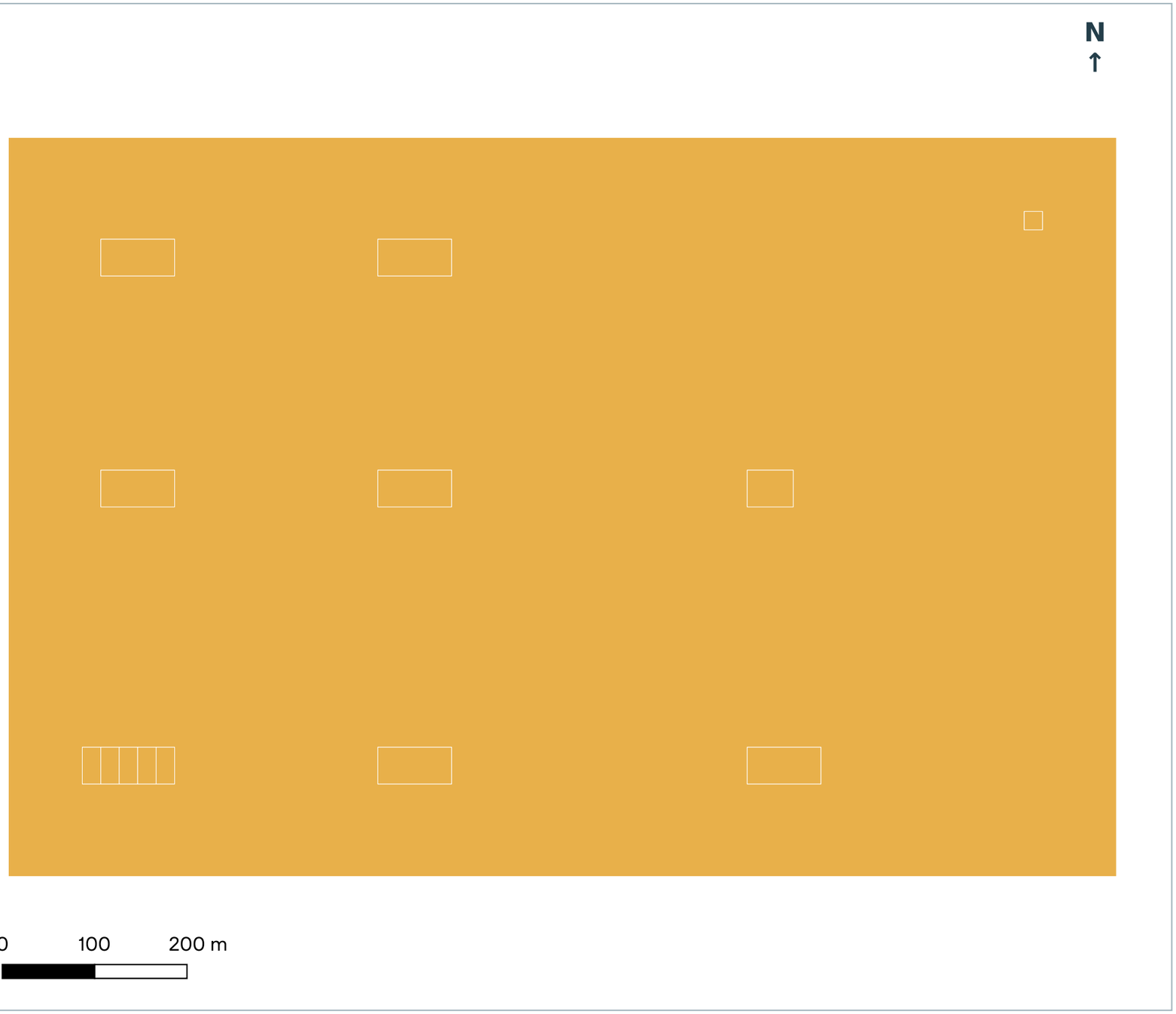
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Tree pit / trench · lined

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated.
FAIL takes precedence over
conditional and admissible.

Each feature stores the reasons,
rule IDs, references, evidence
and required closure actions.

Admissible means screening only.

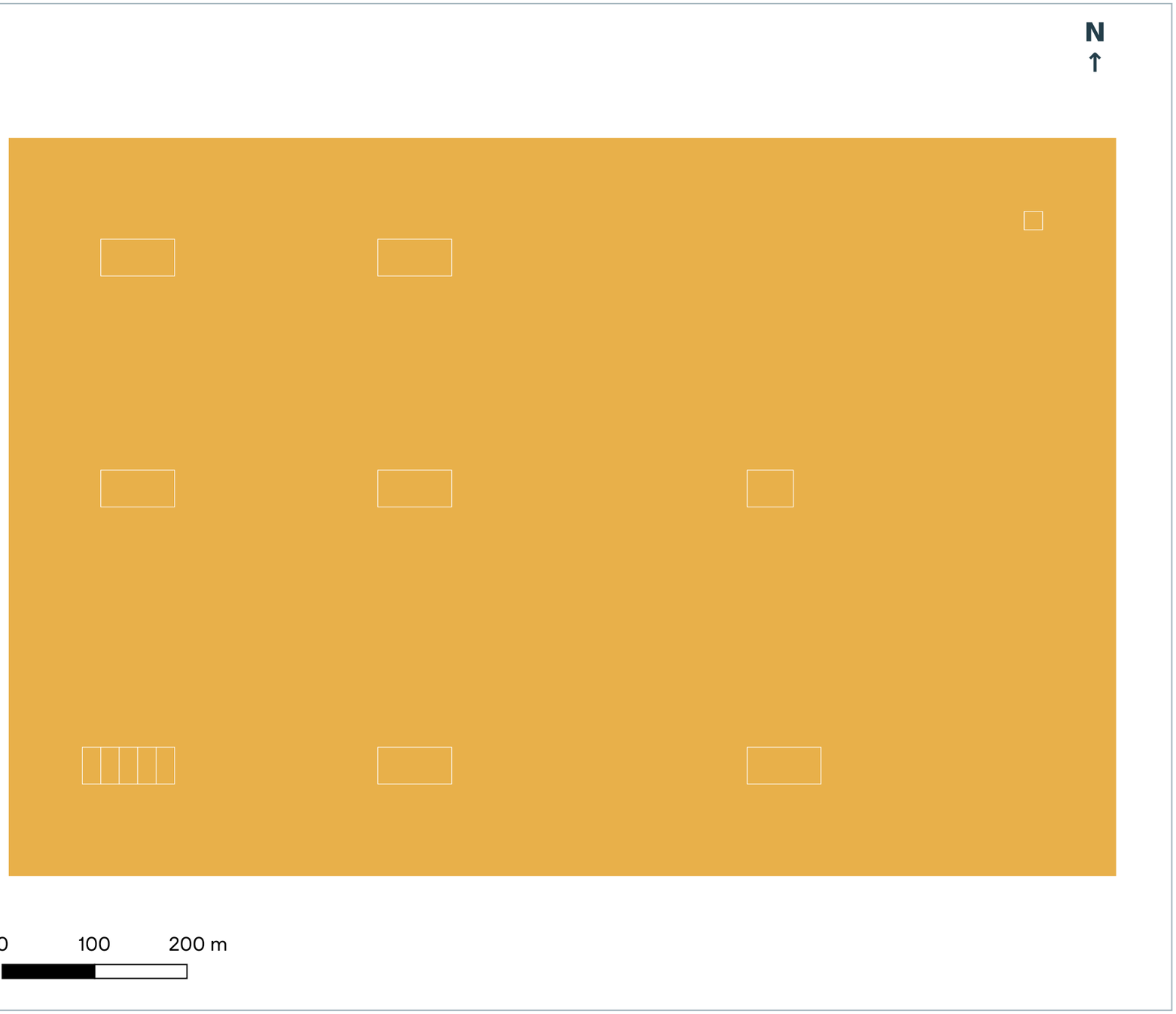
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Permeable pavement · infiltration

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated.
FAIL takes precedence over
conditional and admissible.

Each feature stores the reasons,
rule IDs, references, evidence
and required closure actions.

Admissible means screening only.

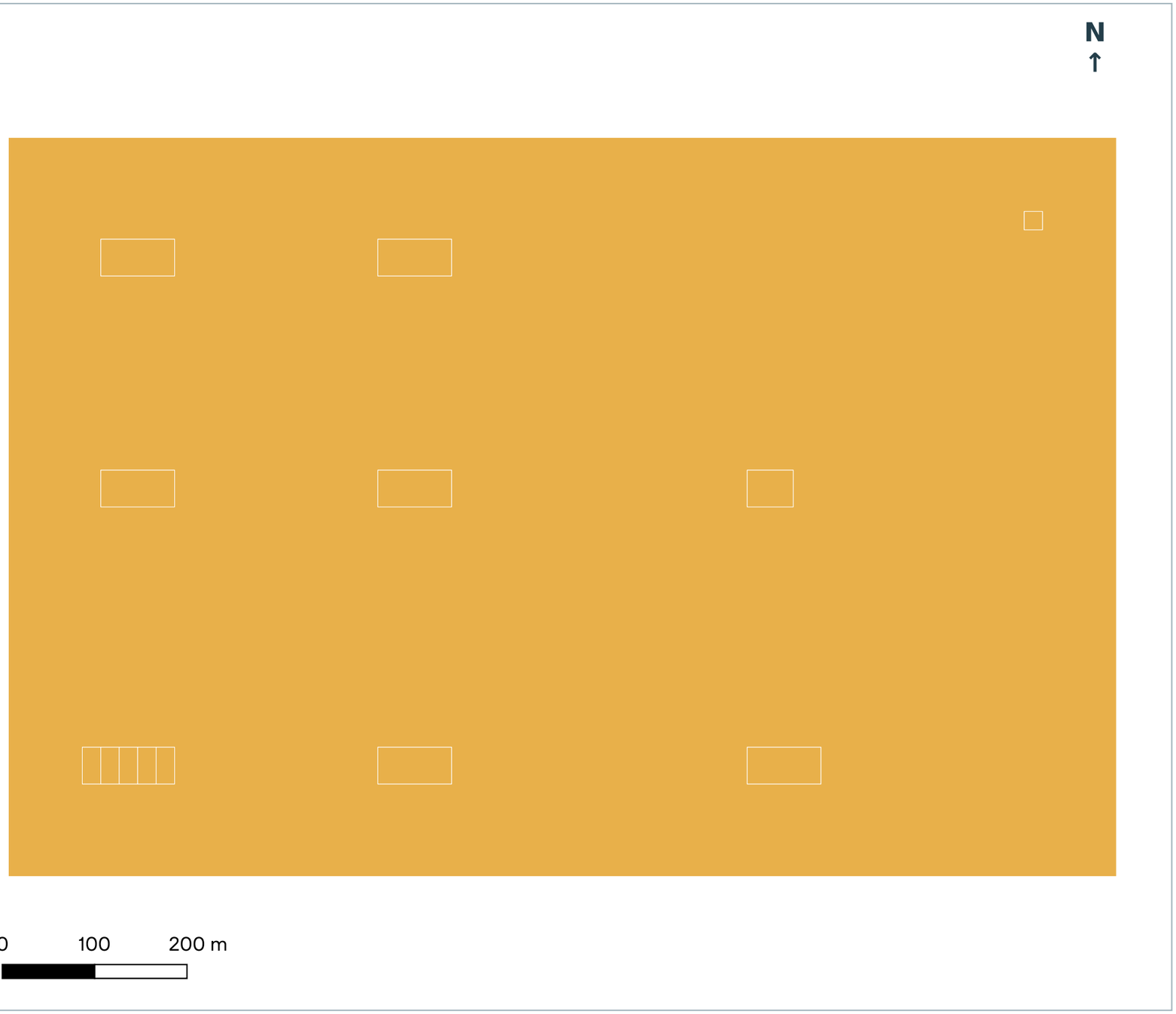
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Permeable pavement · lined / underdrained

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

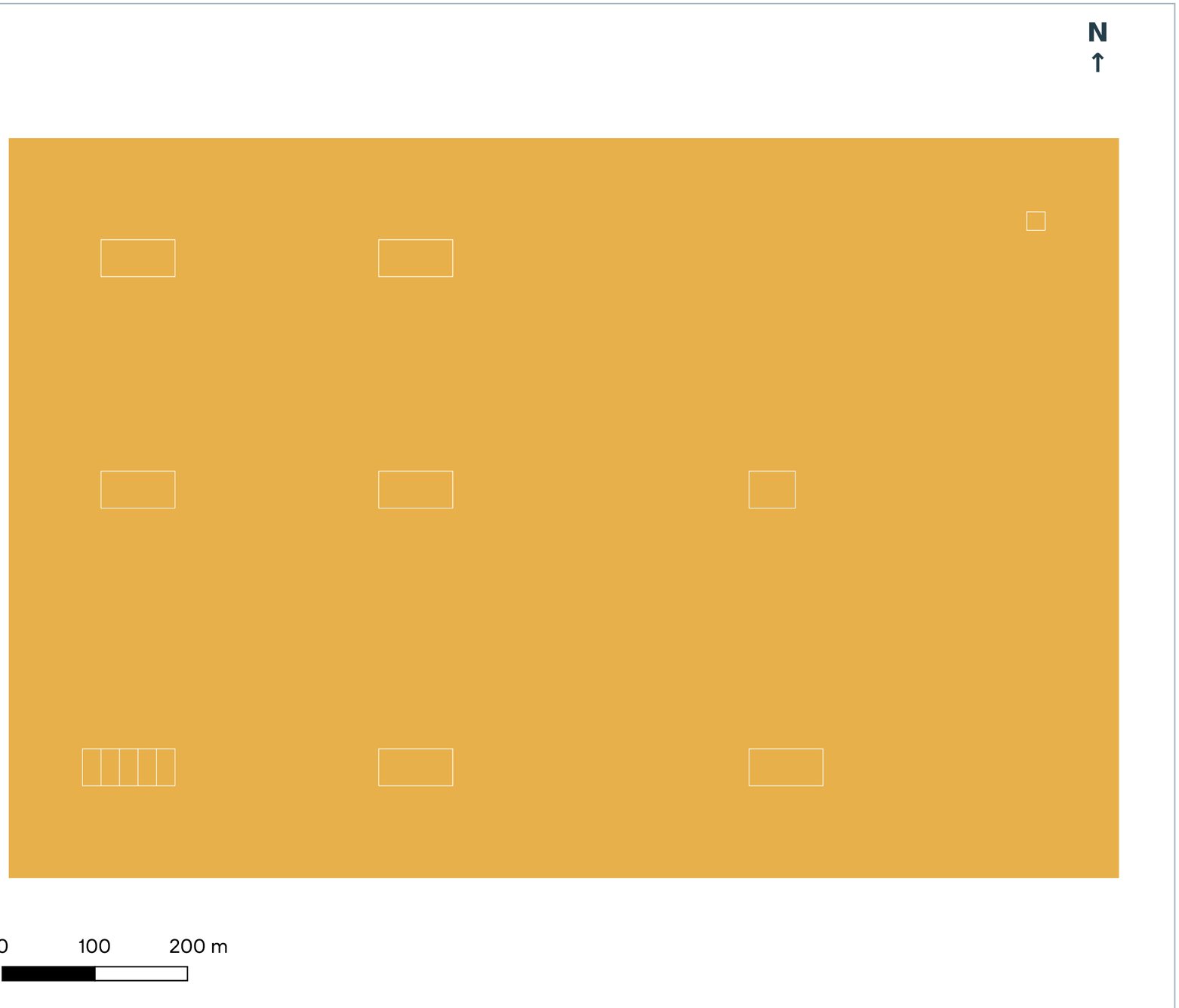
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Subsurface infiltration trench

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

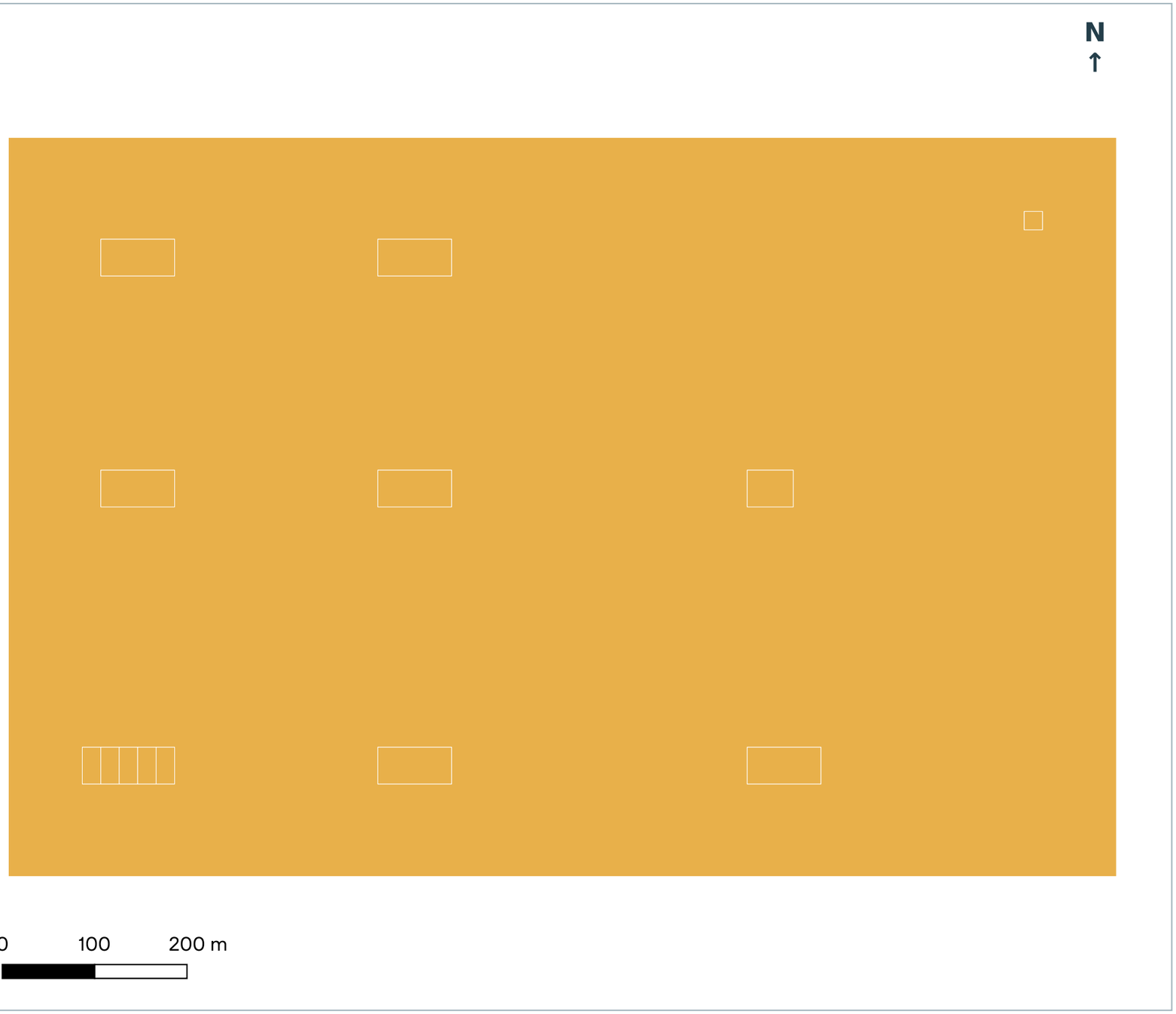
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Combined swale–trench system

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

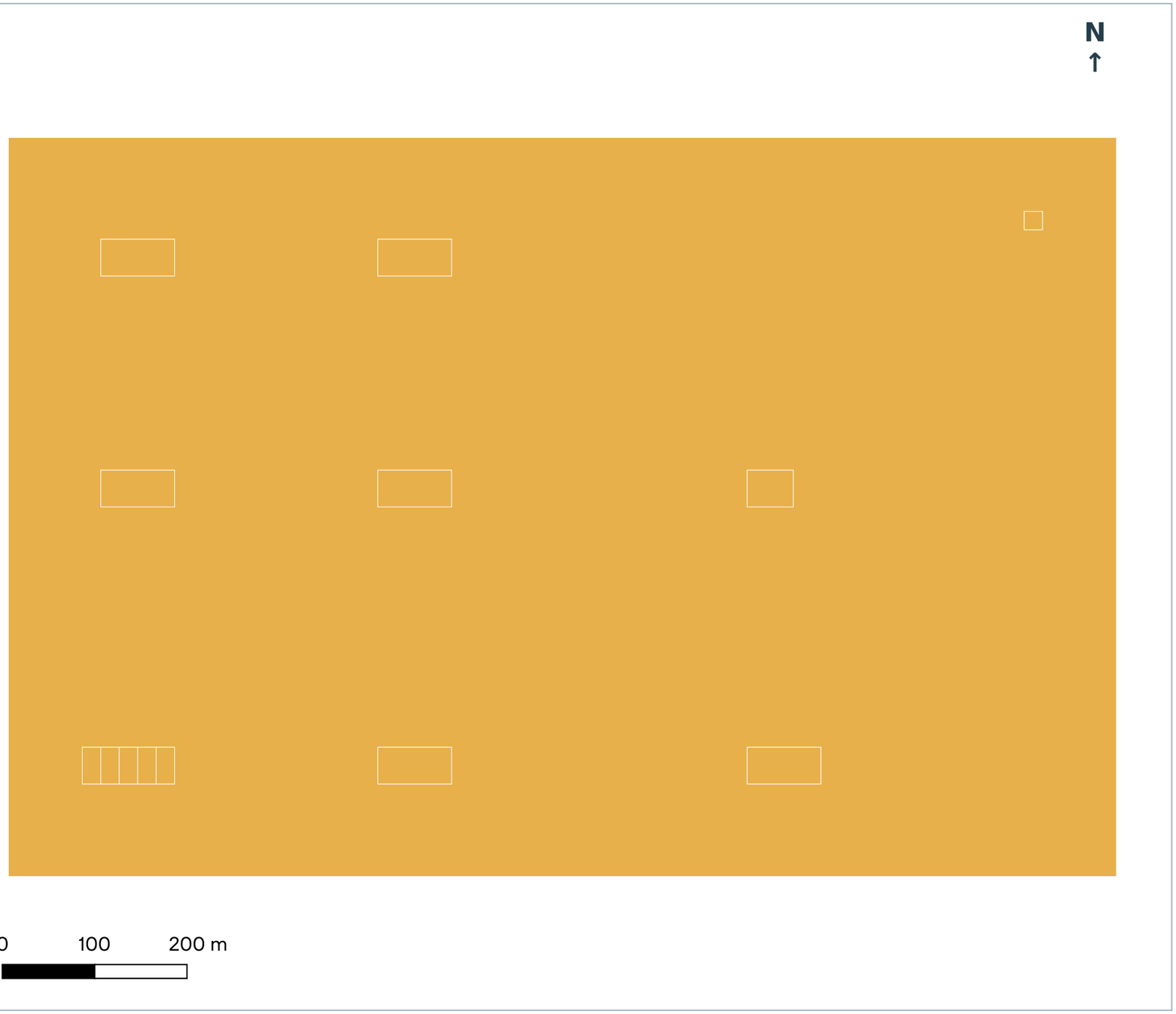
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Infiltration shaft

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

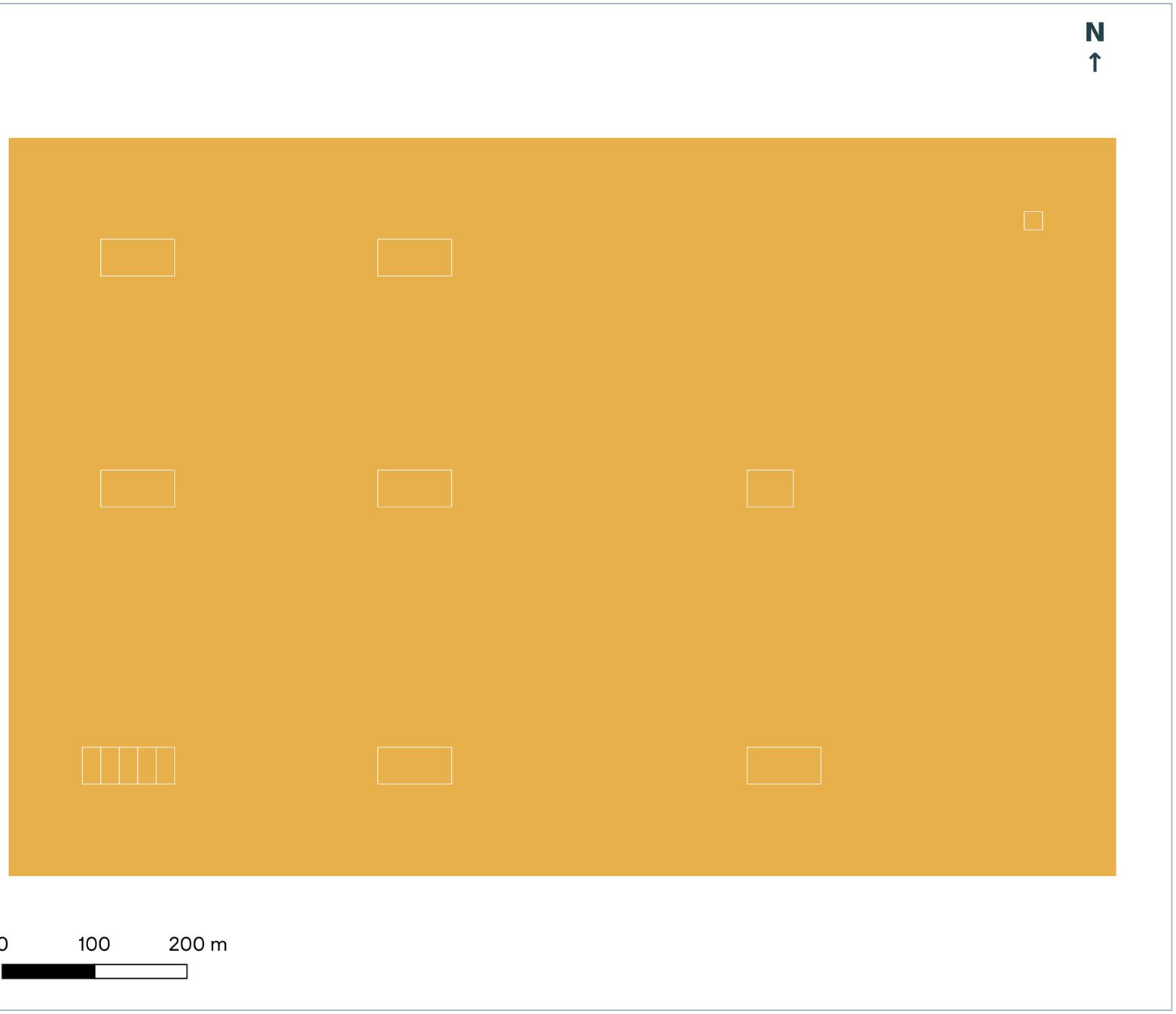
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Lined multifunctional detention

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

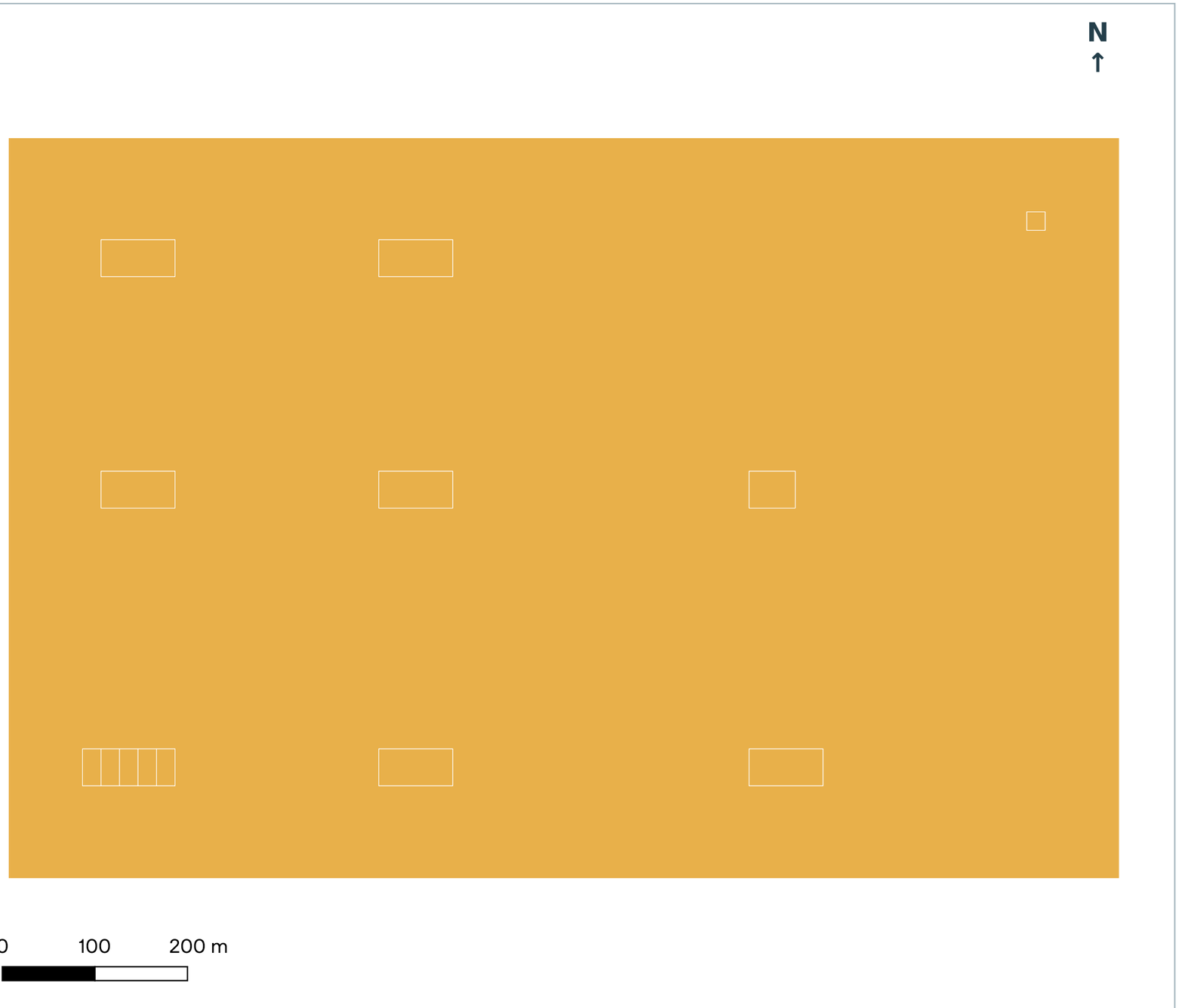
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Safe surface conveyance

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Conditional · evidence / decision needed

READ THIS MAP

All applicable gates are evaluated. FAIL takes precedence over conditional and admissible.

Each feature stores the reasons, rule IDs, references, evidence and required closure actions.

Admissible means screening only.

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · registered flood-depth mask

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Inside registered MURIEL depth mask
- Outside registered MURIEL depth mask
- Unknown source depth / NoData

READ THIS MAP

Area >2000 m²; max depth
>0.3 m. Receptor <1 m;
municipal non-road land <50 m;
usable destination ≤100 m.
Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2,
5.5.3. Reasons and evidence are
stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · flood hotspot screen

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Area and maximum-depth criteria met
- Below local hotspot screen

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

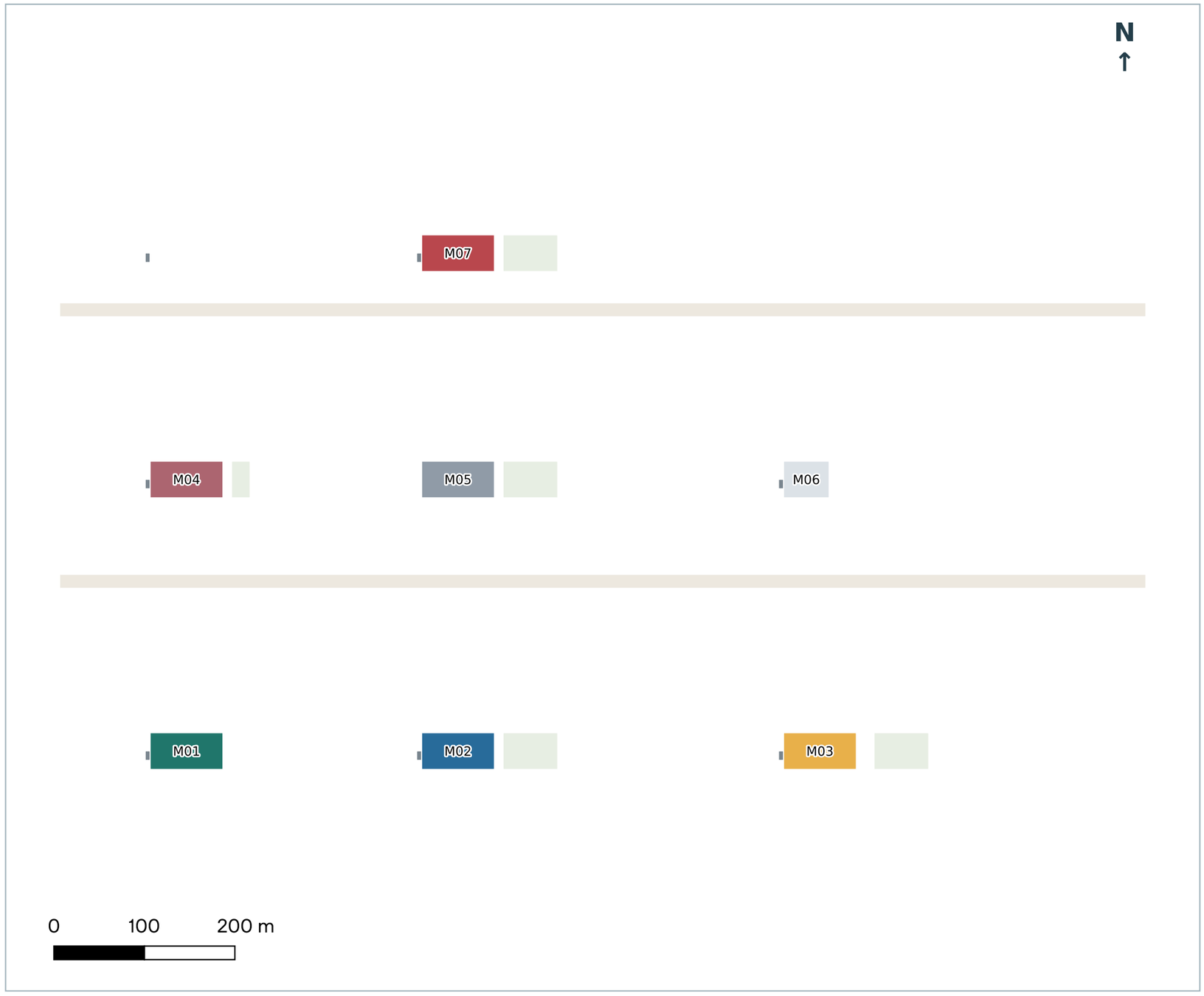
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · protect, investigate or adapt

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Below local hotspot screen
- Protect existing retention
- Retrofit · route evidence supplied
- Investigate topography / flow route
- No usable destination · alternatives
- Outside local adjacency / land screen
- Assessed route incompatible

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors. Pale blue: flood footprint. Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · usable municipal destinations

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Usable municipal non-road space

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · routing evidence and gaps

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Route evidence supplied
- Incompatible assessed route
- Proximity only · no flow proof

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

Dashed lines show proximity or incomplete evidence; no flow proof.

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.